

Here is a brainstormed list of 15 "How might we..." statements designed to help Panasonic Avionics Corporation (PAC) achieve its strategic objectives, overcome challenges, and accelerate its transition to a cloud-native services provider.

1. How might we help them bypass the aviation technician shortage and reduce aircraft downtime (AOG)?

- **Concrete Action:** Implement zero-downtime, asynchronous automation that integrates with ServiceNow to trigger IT service management and MRO configuration changes the exact moment an approval is granted 1, 2.
- **Primary Red Hat Technology:** Red Hat Event-Driven Ansible (EDA) 1, 2.

2. How might we help them escape ballooning legacy virtualization costs post-Broadcom acquisition?

- **Concrete Action:** Execute a VM-to-container lift-and-shift migration, moving existing virtual machines off expensive legacy environments and onto a unified platform that can run both VMs and containers side-by-side 3, 4.
- **Primary Red Hat Technology:** Red Hat OpenShift Virtualization 3, 4.

3. How might we help them ensure flawless software updates to the Astrova seat-end displays in disconnected aircraft environments?

- **Concrete Action:** Package the operating system as an immutable, tamper-proof container so that if an update fails in the field, it can be instantly rolled back like a smartphone, virtually eliminating deployment downtime 5, 6.
- **Primary Red Hat Technology:** RHEL 10 Image Mode and Red Hat Device Edge 5, 6.

4. How might we help them future-proof their network cryptography against emerging threats on new LEO satellite networks?

- **Concrete Action:** Enable out-of-the-box, FIPS-compliant quantum cryptography at install time to secure the massive volumes of passenger and operational data flowing across new multi-orbit satellite networks 7, 8.
- **Primary Red Hat Technology:** Red Hat Enterprise Linux 10 8.

5. How might we help them maintain critical security waivers for vulnerable, aging applications?

- **Concrete Action:** Isolate and containerize legacy Java applications that are currently running on unsupported CentOS 7 platforms to securely harden them and help PAC pass strict internal CrowdStrike cybersecurity audits 9.
- **Primary Red Hat Technology:** Red Hat OpenShift and RHEL Universal Base Image (UBI) 9.

6. How might we empower them to deliver localized, AI-driven passenger personalization directly on the aircraft?

- **Concrete Action:** Serve predictive AI models and generative passenger engagement engines locally at the aircraft edge, providing real-time personalized content recommendations without requiring a constant satellite cloud connection 10, 11.
- **Primary Red Hat Technology:** Red Hat OpenShift AI and Red Hat Device Edge 11.

7. How might we help them overcome technical debt caused by a fragmented "roll-your-own" engineering culture?

- **Concrete Action:** Establish a single "golden path" for Configuration as Code (CaC) and enforce strict DevOps guardrails by integrating deployment automation directly with GitLab 5, 12.
- **Primary Red Hat Technology:** Red Hat Ansible Automation Platform 12.

8. How might we help them resolve Configuration Item (CI) duplication and dynamic inventory sprawl?

- **Concrete Action:** Integrate network automation webhooks with NetBox so that adding a new device or changing a configuration context instantly provisions the infrastructure and natively updates their single source of truth 1, 13.
- **Primary Red Hat Technology:** Event-Driven Ansible (EDA) 13.

9. How might we help them modernize their highly locked-down, air-gapped Media Content Services?

- **Concrete Action:** Automate the secure delivery of media content (movies, music, catalogs) into restricted flight environments by importing data and updates through isolated, controlled pipelines 14.
- **Primary Red Hat Technology:** Red Hat Ansible Automation Platform and Red Hat Satellite 14.

10. How might we help them standardize the provisioning of massive global F5 load balancer infrastructures?

- **Concrete Action:** Replace manual load balancer administration by deploying modular playbooks to automatically provision F5 Big IP nodes, route domains, VLANs, and self-IPs across their core global sites 1, 13.
- **Primary Red Hat Technology:** Red Hat Ansible Automation Platform 13.

11. How might we help them close the technical skills gap and speed up system troubleshooting for administrators?

- **Concrete Action:** Deploy natural-language AI assistants to help newer system administrators query logs, troubleshoot broken services (like SSHD), and rapidly generate automation scripts 15, 16.
- **Primary Red Hat Technology:** RHEL 10 Command Line Assist and Red Hat Lightspeed 16.

12. How might we provide cloud-agnostic portability for their refactored applications to avoid vendor lock-in?

- **Concrete Action:** Build a uniform container platform allowing PAC developers to effortlessly write code once and migrate workloads between Microsoft Azure, AWS, and on-premises environments 17, 18.
- **Primary Red Hat Technology:** Red Hat OpenShift 17, 18.

13. How might we help them eradicate fleet-wide vulnerabilities rapidly and without manual intervention?

- **Concrete Action:** Expand automation licensing across PAC's 5,000+ host inventory to enable enterprise-wide, automated vulnerability remediation and patch management directly from dynamic inventories 19.
- **Primary Red Hat Technology:** Red Hat Ansible Automation Platform 19.

14. How might we help them simplify OS image creation across their hybrid cloud footprint?

- **Concrete Action:** Transition away from maintaining complex, manual golden images by defining the OS via DevOps pipelines, utilizing AI to recommend necessary packages and secure configurations during the build process 20, 21.
- **Primary Red Hat Technology:** RHEL 10 Image Builder 21.

15. How might we assist them in achieving strict EASA Part-IS cybersecurity mandates by 2026?

- **Concrete Action:** Standardize on a secure-by-design, hardened Linux distribution for all next-generation IFEC platforms to guarantee traceability, long-term support, and compliance for their operational Information Security Management System (ISMS) 22, 23.
- **Primary Red Hat Technology:** Red Hat Enterprise Linux (RHEL) 22.